

PDF SLIDE PRESENTATION

Accelerate Your Contributions

Enter a GitHub issue URL to get a comprehensive implementation plan and dependency analysis.

Signed in. New analyses and Ask Bob replies will be saved to Neon.

BOBOPENSOURCE

Accelerating Open Source Contributions with AI-Guided Issue Analysis

A developer workflow platform that reads GitHub issues, analyzes repository structure, maps dependencies, guides implementation, and supports authenticated, persistent collaboration through the web.

Start Analysis

PRESENTATION FOCUS

Product story using the **STAR framework**

Platform stacks and technologies

AI workflow includes **IBM Bob** usage in project delivery

Live demo: **<https://bobopensource-live.vercel.app>**

Developer profile and delivery context

What BobOpenSource Is

Core Product Vision

BobOpenSource is an **AI-powered GitHub issue analyzer** built to reduce the friction developers face when contributing to unfamiliar codebases.

- Reads GitHub issues and issue discussion.
- Analyzes repository structure, conventions, and dependencies.
- Generates implementation roadmaps, test guidance, and risk notes.
- Supports follow-up Q&A through an Ask Bob experience.

Business Value

- Shortens time-to-first-contribution for open source developers.
- Turns vague GitHub issues into actionable engineering guidance.
- Improves confidence through risk analysis, code hints, and validation steps.
- Persists analyses and conversations for repeatable team workflows.
- Provides a live web experience at **bobopensource-live.vercel.app**.

BobOpenSource combines repository intelligence, issue understanding, and guided execution into one developer-facing experience.

INPUTS

Issue URL + Repo

OUTPUTS

Plan + Q&A

MODES

CLI + Web

PERSISTENCE

Auth + DB

Situation

SITUATION

TASK

ACTION

RESULT

Problem Space

- Developers often want to contribute to open source but lack fast context.
- GitHub issues may be vague, broad, or disconnected from repository structure.
- New contributors need to know where to start, which files matter, and how to test changes.

Operational Gaps

- Repository onboarding is manual and time-consuming.
- Dependency and impact analysis is usually done by hand.
- Planning often becomes generic rather than grounded in actual codebase context.

The situation called for a tool that could reason about both the GitHub issue and the repository itself.

Task

SITUATION

TASK

ACTION

RESULT

Primary Task

Design and implement a platform that translates a GitHub issue into a **repository-aware implementation plan**.

- Identify relevant files and components.
- Recommend code changes and validation steps.
- Expose the experience in both CLI and web interfaces.

Secondary Task

- Add an interactive Ask Bob layer for follow-up developer questions.
- Persist user analyses and responses behind authentication.
- Make the guidance concrete enough to help a contributor start coding with confidence.

Action

SITUATION

TASK

ACTION

RESULT

Platform Actions

- Built a TypeScript analysis engine for repository scanning, issue parsing, dependency mapping, and implementation planning.
- Added a Next.js web interface with guided tabs for insights, dependencies, plan, and Ask Bob.
- Introduced repository-aware code hints, diff previews, risk summaries, and PR draft generation.

Product Actions

- Improved remote GitHub analysis with file ranking, README context, and issue discussion capture.
- Integrated Clerk for GitHub-based login and route protection.
- Integrated Neon for saved analyses and Ask Bob conversation persistence.

The action phase focused on making AI guidance useful, inspectable, and durable for real contributors.

Result

SITUATION

Contributor Guidance

- Issue complexity and effort estimation
- Suggested files and code-path hints
- Step-by-step implementation roadmap

TASK

Execution Support

- Suggested code and test scaffolds
- Validation checklists and risk notes
- PR draft output for contributor handoff

ACTION

RESULT

Product Outcomes

- Authenticated user experience with Clerk
- Saved analyses and chats in Neon
- Interactive Ask Bob workflow for follow-up questions

Result: BobOpenSource now operates as a guided contribution platform, not just a static analyzer.

Stacks and Tools Used

Core Engine

- TypeScript
- Node.js
- Commander CLI
- Simple Git
- Glob and YAML parsing

Web Platform

- Next.js 16 App Router
- React 19
- Framer Motion
- Lucide React
- TailwindCSS 4

Auth and Data

- Clerk for GitHub login
- Neon Serverless Postgres
- GitHub API integration
- JSON persistence of analyses and chats

Quality

- TypeScript strict mode
- Jest
- ESLint
- Prettier

Key Product Capabilities

- Issue parsing
- Repository analysis
- Dependency mapping
- Ask Bob contextual Q&A

Deployment Readiness

- Web + CLI separation
- Protected APIs
- Persistent user records
- Presentation-ready product story

BobOpenSource Workflow

Processing Flow

- Input GitHub issue URL
- Analyze repository locally or remotely
- Parse issue requirements and discussion
- Build dependency map
- Generate roadmap, test strategy, and risks
- Support follow-up questions through Ask Bob

Persistence Flow

- Clerk authenticates the user with GitHub
- Analyze API saves issue analysis into Neon
- Ask Bob API stores Q&A records linked to the analysis
- Future extension: saved history and reusable contribution workspaces

The architecture is designed to turn one-off analysis into a reusable contributor knowledge trail.

Daniel Mwihoti

Role

Software Developer

Built and evolved BobOpenSource as a developer-focused platform for issue interpretation, implementation guidance, and authenticated knowledge capture.

- Product thinking around developer onboarding workflows
- Full-stack delivery across CLI, web, auth, and persistence
- Focus on practical AI assistance rather than generic automation
- Used IBM Bob in the project workflow: <https://bob.ibm.com/>

Professional Links

- **LinkedIn:** [linkedin.com/in/daniel-mwihoti-3aaa652b9](https://www.linkedin.com/in/daniel-mwihoti-3aaa652b9)
- **GitHub:** github.com/mwihoti
- **Project Repo:** github.com/mwihoti/bobopensource
- **Live App:** bobopensource-live.vercel.app
- **IBM Bob:** bob.ibm.com

Presentation Summary

- Product: BobOpenSource
- Format: PDF slide presentation
- Method: STAR storytelling
- Development support: IBM Bob
- Deployment: <https://bobopensource-live.vercel.app>
- Audience: product, engineering, portfolio, or demo stakeholders